

Random Enough: A Connectome Test of the Fly Hashing Algorithm

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Abstract—The *Drosophila* olfactory circuit has been characterised as a locality-sensitive hash: a sparse binary expansion from ~ 50 glomeruli onto ~ 2000 Kenyon cells, followed by winner-take-all sparsification, which outperforms classical locality-sensitive hashing (LSH) at similarity search. Because no reconstruction of the circuit existed when that result was obtained, its projection matrix was generated at random, anchored to biology only through the number of inputs each Kenyon cell receives. The MaleCNS connectome now permits the substitution of measured connectivity for that idealisation. Evaluating nearest-neighbour retrieval over measured odorant responses, we find the architectural advantage intact ($4.8\times$ over classical LSH), while the measured wiring performs marginally *worse* than a curveball null that fixes both degree sequences and randomises only the pairing: a 90% bootstrap interval over odorants places the difference between -2.2% and -0.3% at the largest hash size. The protocol resolves effects of 1.7% , and rejects a deliberately degraded matrix at $z < -100$, so the interval reflects the circuit rather than the instrument. We do not claim the wiring is unstructured, which independent work has refuted; we claim that the structure it carries does not serve this computation, and costs a little of it.

Index Terms—connectomics, locality-sensitive hashing, similarity search, neural architectures, null models

I. INTRODUCTION

Similarity search — retrieving the database items most like a query — is a primitive of information retrieval, and locality-sensitive hashing (LSH) is its standard sublinear solution [1], [2]. An LSH maps items to short codes under which similar items collide with high probability, so that expensive comparisons in the input space are replaced by cheap comparisons over codes.

Dasgupta, Stevens and Navlakha [3] observed that the fruit fly’s olfactory circuit computes something structurally analogous, and that it departs from classical practice in three respects: its projection is sparse and binary rather than dense and Gaussian, it expands dimensionality rather than reducing it, and it sparsifies the expanded representation by winner-take-all. Evaluated on standard retrieval benchmarks, this “fly algorithm” improved on classical LSH by as much as $3\times$.

The projection matrix in that work was necessarily synthetic. Connectivity between olfactory projection neurons and Kenyon cells had not been reconstructed, so the authors sampled a sparse binary random matrix whose sole biological anchor was a mean of roughly six inputs per Kenyon cell, consistent with anatomical estimates. The choice was well motivated: the prevailing account held such connectivity to be effectively random [4]. Whether the idealisation was *accurate*, as opposed to merely convenient, could not be determined.

Two developments make the question answerable. Zheng *et al.* [5] exploited the hemibrain volume to demonstrate that Kenyon cells sample olfactory input in a measurably non-random fashion, displacing the earlier account. Independently, the MaleCNS connectome [6] supplies a complete synaptic reconstruction of an adult central nervous system, and with it the entire mushroom body input layer of a single animal.

Taken together these results pose a question that neither resolves. The wiring is structured, and the circuit is argued to implement a hash; it does not follow that the structure is *for* the hash. Substituting measured connectivity for the idealisation of [3] therefore tests a specific functional hypothesis rather than a general claim about randomness. We report that the substitution yields no benefit, and a small reliable cost. Our contributions are:

- the first evaluation of the fly hashing algorithm under measured rather than sampled connectivity;
- a null-model protocol that separates the contribution of wiring *pairing* from the degree sequences with which it is confounded, paired with calibration matrices that establish the effect size the protocol can resolve;
- an equivalence-test formulation, with uncertainty propagated at the level of odorants rather than queries, under which the measured matrix is bounded within 2.2% of its null — on the deficit side.

II. BACKGROUND

A. Locality-Sensitive Hashing

A family $\mathcal{H}$ of hash functions over a metric space $(\mathcal{X}, D)$ is (r, cr, p_1, p_2) -sensitive if, for h drawn uniformly from $\mathcal{H}$,

$$D(u, v) \leq r \Rightarrow \Pr[h(u) = h(v)] \geq p_1, \quad (1)$$

$$D(u, v) \geq cr \Rightarrow \Pr[h(u) = h(v)] \leq p_2, \quad (2)$$

with $c > 1$ and $p_1 > p_2$ [1]. Classical constructions reduce dimensionality with dense random projections and binarise, e.g. the sign of a Gaussian projection [2].

B. The Fly Circuit as a Hash

Olfactory receptor neurons converge onto $d \approx 51$ glomeruli, whose projection neurons diverge onto $m \approx 2000$ Kenyon cells. Two mechanisms shape the code. Divisive normalisation in the antennal lobe removes concentration, retaining odour identity [7]. A single giant inhibitory neuron, APL, implements approximate winner-take-all, leaving a few percent of Kenyon cells active [8], [9]. The resulting sparse binary pattern

is the circuit's tag for an odour. The same construction also supports novelty detection [10].

C. Null Models for Connectomes

Asking whether a measured network is “special” requires a null preserving everything uninteresting. Degree-preserving randomisation [11] is standard: it retains each node's connection count while destroying which node connects to which. In a bipartite projection, two degree sequences may be preserved independently, and which one is held fixed changes the question being asked.

III. METHODS

A. The Measured Projection

Let $M \in \{0,1\}^{m \times d}$ with $M_{ij} = 1$ iff glomerulus j synapses onto Kenyon cell i . Define the *claw count* of cell i and the *fan-out* of glomerulus j as

$$c_i = \sum_j M_{ij}, \quad f_j = \sum_i M_{ij}. \quad (3)$$

We build M from MaleCNS v1.0 [6] using every reconstructed connection, imposing no synapse-count threshold: the resulting graph has 2.56×10^7 connections and 1.24×10^8 synapses. Thresholding is common when connectomes drive dynamical models [12], but Kenyon cell claws are individually weak, and a threshold removes them: at a ≥ 5 synapse cut the mean claw count falls to 4.79, against 5.28 with no cut and an anatomical expectation near six.

Projection neurons are identified by the ALPN class and assigned to glomeruli from their cell-type labels; multiglomerular and unnamed types are excluded, as are thermo- and hygro-sensory channels. Restricted to one hemisphere this yields $d = 51$ glomeruli and $m = 1886$ Kenyon cells.

B. The Hash

Following [3], for an input $x \in \mathbb{R}_{\geq 0}^d$ of glomerular responses:

$$\tilde{x} = \frac{d}{\sum_j x_j} x \quad (\text{divisive normalisation}), \quad (4)$$

$$y = M\tilde{x} \quad (\text{sparse binary expansion}), \quad (5)$$

$$h_k(x)_i = \mathbf{1}[y_i \geq y_{(k)}] \quad (\text{winner-take-all}), \quad (6)$$

where $y_{(k)}$ is the k -th largest entry of y . The tag $h_k(x) \in \{0,1\}^m$ has exactly k ones. Tags are compared by Hamming distance.

The classical baseline draws $W \in \mathbb{R}^{d \times k}$ with i.i.d. Gaussian entries and returns $\text{sign}(\tilde{x}^\top W)$, so both methods expend k bits.

C. Null and Calibration Models

Only M varies across conditions; input, normalisation, sparsity and scoring are held fixed, so any difference is attributable to connectivity alone.

- $\mathcal{N}_{\text{claw}}$ preserves $\{c_i\}$ and draws each cell's inputs uniformly without replacement, thereby flattening fan-out.

- $\mathcal{N}_{\text{conf}}$ preserves *both* $\{c_i\}$ and $\{f_j\}$ and randomises only the pairing. This is the null of record: a difference it exposes cannot be ascribed to either degree sequence.
- $\mathcal{N}_6$ reproduces the construction of [3], six uniformly drawn inputs per cell.

Sampling $\mathcal{N}_{\text{conf}}$ correctly requires care. Permuting edge endpoints and discarding the resulting collisions is attractively simple but loses several percent of the connections, and therefore samples from the wrong space; the deficit systematically handicaps the null. We instead use curveball swaps [14]: two Kenyon cells are drawn, the glomeruli unique to each are pooled, and the pool is redealt between them at the original cardinalities. Both margins are invariant by construction, which we verify exactly, and 30 attempted swaps per cell is well beyond the mixing time at this density.

Two further matrices are not nulls but calibration probes, included to establish what the protocol can detect. $\mathcal{P}_{\text{bal}}$ preserves $\{c_i\}$ while equalising fan-out as far as the claw counts permit, and is expected to be mildly superior; $\mathcal{P}_{\text{deg}}$ restricts every cell to the same four glomeruli, and is expected to be far worse. Without such probes a null result is uninterpretable, since an insensitive test and a true absence of effect produce identical output.

D. Evaluation

For query q let $\text{NN}_\kappa(q)$ be its κ nearest items by Euclidean distance in $\mathbb{R}^d$ and $R_\kappa(q)$ the κ retrieved by Hamming distance on tags, ranked. With $\text{rel}(i) = \mathbf{1}[R_\kappa(q)_i \in \text{NN}_\kappa(q)]$,

$$\text{AP}_\kappa(q) = \frac{1}{\kappa} \sum_{i=1}^{\kappa} \text{rel}(i) \cdot \frac{\sum_{l \leq i} \text{rel}(l)}{i}, \quad (7)$$

and mAP averages over queries. We use $\kappa = 10$.

Two sources of variation must be carried. Retrieval quality varies across draws of a null matrix, and it varies across the items queried; and because queries are constructed from a smaller set of measured odorants, they are not independent. We therefore resample *odorants*, not queries: for each of 2000 bootstrap draws we sample odorants with replacement, restrict to the queries they dominate, and recompute the difference between the measured matrix and the mean over $B = 40$ null draws.

Let Δ denote that difference and $\bar{s}_0$ the null mean. We report the 90% interval of $\Delta/\bar{s}_0$, and declare equivalence at margin δ when the interval lies within $(-\delta, +\delta)$, which is the two-sided-tests criterion. This matters: a non-significant difference is not evidence of absence, whereas an interval bounded inside $\pm\delta$ is evidence that no effect larger than δ is present. We also report $z = (s(M_{\text{real}}) - \bar{s}_0)/\sigma_0$ for comparability with the connectomics literature, and the two-sigma resolution $\delta_{2\sigma} = 2\sigma_0/\bar{s}_0$.

E. Data

Odorant responses come from DoOR 2.0 [13], mapped from receptor to glomerulus. Its consensus matrix is $\sim 17\%$ populated, so we retain odorants measured across at least 12 glomeruli and glomeruli measured across at least 40 odorants,

TABLE I
MEAN AVERAGE PRECISION AT $\kappa = 10$, $n = 4000$, $B = 40$.

k	real	$\mathcal{N}_{\text{claw}}$	$\mathcal{N}_{\text{conf}}$	$\mathcal{N}_6$	LSH
8	0.1965	0.2102	$0.2051 \pm .0077$	0.2469	0.0182
16	0.2882	0.3015	$0.2940 \pm .0055$	0.3065	0.0662
32	0.3531	0.3699	$0.3581 \pm .0034$	0.3535	0.1514
64	0.3960	0.4137	$0.4009 \pm .0035$	0.3847	0.2462

TABLE II
EFFECT SIZE AGAINST $\mathcal{N}_{\text{CONF}}$, AND WHAT THE PROTOCOL RESOLVES. THE INTERVAL IS A 90% BOOTSTRAP OVER ODORANTS.

k	$\Delta/\bar{s}_0$ CI	$\delta_{2\sigma}$	$z(\mathcal{P}_{\text{bal}})$	$z(\mathcal{P}_{\text{deg}})$
8	$[-6.9, -1.2]\%$	7.5%	+0.27	-26.1
16	$[-3.5, -0.4]\%$	3.7%	+0.43	-52.9
32	$[-2.6, -0.2]\%$	1.9%	+1.40	-104.0
64	$[-2.2, -0.3]\%$	1.7%	+1.39	-113.3

giving 172 odorants over 35 glomeruli; the projection is restricted accordingly to 35×1838 . Because 172 items leave the null too broad to resolve small effects, queries are Dirichlet-weighted blends of two to five measured odorants — mixtures being what a fly actually encounters — yielding $n = 4000$ items that retain the natural correlation structure.

IV. RESULTS

A. The Architecture Reproduces Its Advantage

Table I and Fig. 1 (left) show the expansion and winner-take-all construction outperforming the Gaussian baseline at an equal bit budget by a factor of 4.8 averaged over hash sizes, qualitatively reproducing [3]. The reproduction is in kind rather than in magnitude, and deliberately so: the measured matrix has fixed dimensions, so the benchmarks of [3] cannot be passed through it without first projecting them into the circuit’s input space, which would no longer test the circuit. Our evaluation is therefore conducted at the fly’s own scale and in its own modality, where the comparison is meaningful.

B. The Measured Wiring Confers No Advantage

Against $\mathcal{N}_{\text{conf}}$ the measured matrix yields $z = -1.11, -1.04, -1.47, -1.40$ at $k = 8, 16, 32, 64$, with 33, 35, 38, 38 of 40 null draws scoring higher. Under the odorant-level bootstrap the 90% interval on the relative difference is $[-6.9\%, -1.2\%]$, $[-3.5\%, -0.4\%]$, $[-2.6\%, -0.2\%]$ and $[-2.2\%, -0.3\%]$ respectively.

Two things follow, and they should not be conflated. First, the interval excludes zero at every hash size, so the deficit is reliable rather than noise: measured connectivity is slightly worse for retrieval than a degree-matched random rewiring. Second, the deficit is small and bounded — within 2.2% at $k = 64$ — so the measured matrix is equivalent to its null at any margin a practitioner would plausibly care about. The appropriate summary is not that wiring makes no difference, but that it makes a small negative one.

C. Calibration and Resolution

Table II bounds the instrument. A degraded matrix is rejected at z between -26 and -113 , so gross departures are detected without difficulty. An improvement of the magnitude produced by $\mathcal{P}_{\text{bal}}$ reaches $z = +1.40$ and is not resolved, consistent with a two-sigma resolution of 1.7% at $k = 64$. The bootstrap interval is tighter than this threshold because it pools information across queries rather than across matrices, which is why we treat it, and not z , as the primary statistic.

D. Robustness

The conclusion is stable under the choices that might plausibly have driven it. Imposing a ≥ 5 synapse reconstruction threshold reduces the graph fourfold, to 6.24×10^6 connections, and moves the mean claw count from 5.28 to 4.79; z remains within $[-1.82, -1.12]$. The result also holds for the contralateral mushroom body, for synapse-weighted rather than binary M , and for Gaussian, low-rank and sparse synthetic inputs, over which z ranges in $[-0.81, +1.25]$ without consistent sign.

One asymmetry deserves separate record, because it suggests where the deficit originates. Measured fan-out is strongly non-uniform: DA1, the glomerulus carrying the pheromone cVA, drives 473 Kenyon cells, while the least connected drives 19. Both $\mathcal{N}_{\text{claw}}$ and $\mathcal{P}_{\text{bal}}$, which flatten this distribution while leaving claw counts intact, score above the measured matrix at every k . Disproportionate allocation of mushroom body capacity to behaviourally privileged channels is precisely the kind of structure reported by [5], and on this evidence it is paid for in general-purpose retrieval.

V. DISCUSSION

Our result should not be read as a claim that projection-neuron to Kenyon-cell connectivity is random. Zheng *et al.* [5] established that it is not, and the fan-out asymmetry documented above is itself structure, visible in the same data we analyse. The claim is narrower: the structure that is present confers no advantage on similarity search, the computation the circuit has been argued to perform, and in fact exacts a small cost.

This reconciles the two literatures rather than opposing either. The idealisation adopted in [3] was not merely expedient but substantively adequate for its purpose, since with respect to retrieval quality a measured matrix and a degree-matched random one are interchangeable to within a few percent. Where a computation requires a random projection, evolution appears to have supplied something close enough to one; the departures from randomness that do exist are then free to serve other ends, of which innate valence, developmental constraint and the over-allocation visible in fan-out are the obvious candidates. A circuit can implement an algorithm well without its every structural feature being an adaptation for that algorithm, and distinguishing the two requires exactly the kind of null the connectome now makes available.

The methodological point generalises beyond olfaction. Connectome-constrained models are increasingly used to

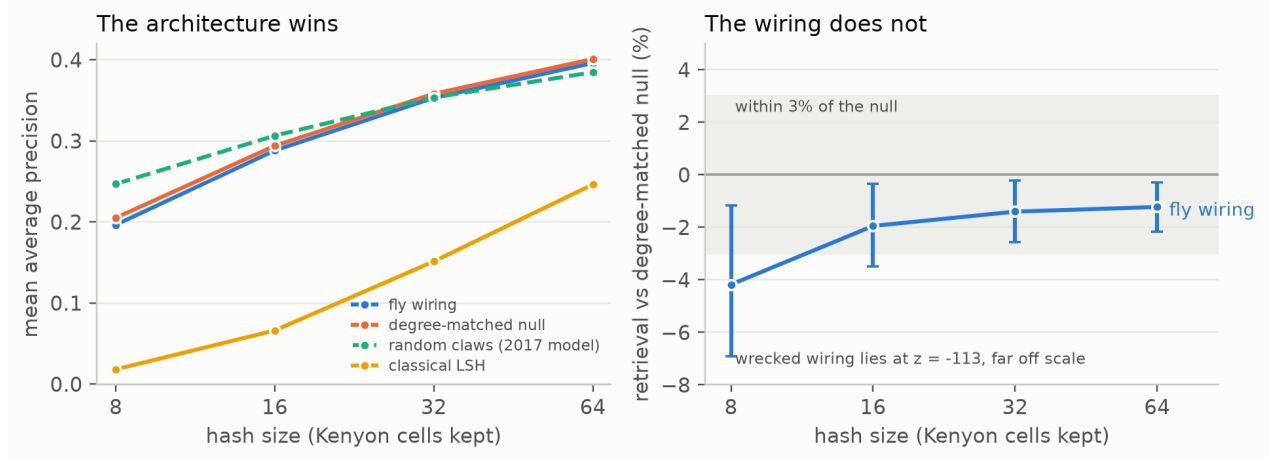


Fig. 1. **Left:** the architectural advantage over classical LSH survives the substitution of measured connectivity; measured matrix, null and the construction of [3] are mutually indistinguishable at this scale. **Right:** retrieval relative to that null, with 90% bootstrap intervals over odorants. The deficit excludes zero at every hash size yet stays within 3% for $k \geq 16$; the degraded probe falls at $z = -113$.

ground computational claims [12], [15], and such claims are testable in a way they recently were not: fix every component of a circuit-derived algorithm except its measured connectivity, randomise that against nulls which preserve the uninteresting invariants, and calibrate the comparison against matrices whose quality is known in advance. Absent the last step, a null result is indistinguishable from an insensitive measurement.

VI. LIMITATIONS

The central claim is a bounded difference rather than an absence: effects smaller than roughly two percent are not excluded, and tightening the bound requires more queries, which is quadratic in the retrieval evaluation. We examine a single task under a single metric; novelty detection [10], the circuit’s other documented function, may weight connectivity differently and is untested here. DoOR covers 35 of 51 glomeruli, so the projection is evaluated on a subset of its input channels, and the 4000 queries are mixtures of 172 measured odorants rather than independent measurements — which the odorant-level bootstrap accounts for in its uncertainty, but which still limits the effective sample. MaleCNS reconstructs one male animal; whether the conclusion holds across individuals, or against the female FlyWire reconstruction [16], remains open, and a cross-sex comparison is now feasible.

VII. CONCLUSION

Replacing the random projection assumed in [3] with measured mushroom body connectivity leaves similarity-search performance within 2.2% of a degree-matched null, on the deficit side, while the architectural advantage over classical LSH is undiminished. The mushroom body is an effective hash function by virtue of its shape rather than its wiring, and the structure its wiring does carry appears to serve some other purpose at a modest cost to this one. Code, data pipeline and results are available at <https://github.com/ssenge/FlyHash-Connectome>.

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